Recommender Systems on the MARS E-Learning Dataset: Collaborative Filtering, Content-Based Filtering, and Matrix Factorisation via SVD

Sourav M B
Roll No: CB.PS.P2ASD25023
MSc Applied Statistics and Data Analytics
Department of Mathematics
School of Physical Sciences
Amrita Vishwa Vidyapeetham, Coimbatore

Kashyap Ramakrishnan
Roll No: CB.PS.P2ASD25016
MSc Applied Statistics and Data Analytics
Department of Mathematics
School of Physical Sciences
Amrita Vishwa Vidyapeetham, Coimbatore

***Abstract*—This report presents a recommender system study conducted on the MARS (Mandarine Academy Recommender System) dataset, a real-world e-learning dataset published on Harvard Dataverse. The dataset contains 85,339 explicit ratings from French-language corporate training videos on Microsoft Office 365 tools. The study was undertaken as Part 2 of the Data Mining course (24ASD513) under the guidance of Dr. P.N. Kumar. Four recommendation approaches were implemented and evaluated: user-based collaborative filtering using cosine similarity, user-based collaborative filtering using Pearson similarity, item-based content-based filtering using TF-IDF on item metadata, and matrix factorisation using Truncated Singular Value Decomposition (SVD). Evaluation metrics included Root Mean Square Error (RMSE), Precision at K, and Recall at K. The SVD model achieved an RMSE of 2.2678 on a 1–10 rating scale. Precision@5 was 0.0708 and Recall@5 was 0.0179 on a sample of 100 test users. A comparison of cosine and Pearson similarity revealed that Pearson produces more discriminative similarity scores for users with rating scale bias, which is a key characteristic of this dataset.**

***Index Terms*—recommender systems, collaborative filtering, content-based filtering, matrix factorisation, SVD, cosine similarity, Pearson similarity, e-learning, MARS dataset, Precision@K, Recall@K**

I. INTRODUCTION

Recommender systems are information filtering systems that predict the preferences of users for items they have not yet interacted with. They are foundational to modern digital platforms and have been studied extensively since the Netflix Prize competition of 2006–2009. The three principal paradigms are collaborative filtering (which models user–user or item–item similarity), content-based filtering (which models item features), and matrix factorisation (which decomposes the user–item interaction matrix into latent factors).

This project implements all three paradigms on a real-world corporate e-learning dataset. The choice of dataset is deliberate: the MARS dataset (Mandarine Academy Recommender System) is one of the few publicly available e-learning recommender datasets with explicit ratings, a domain that is underrepresented in standard benchmarks. Unlike movie or product ratings, e-learning ratings are derived from a behavioural signal (watch percentage), which makes the rating distribution unusual and poses specific modelling challenges.

The objectives of this study are:

- To construct a user–item rating matrix from explicit ratings and characterise its sparsity.
- To implement user-based collaborative filtering using cosine similarity with a graceful fallback for cold-start cases.
- To implement user-based collaborative filtering using Pearson correlation-based similarity and compare its behaviour to cosine similarity.
- To implement content-based filtering using TF-IDF on item metadata (name, difficulty, type).
- To implement matrix factorisation using mean-centred Truncated SVD and evaluate it on a held-out test set.
- To evaluate the SVD model using RMSE, Precision@5, and Recall@5.

II. DATASET DESCRIPTION

A. Source and Licensing

The MARS dataset was published by Farah et al. in 2022 and is hosted on Harvard Dataverse [1]. It was also described in a data paper published in *Data in Brief* (Elsevier) [2]. The dataset is licensed under Creative Commons Attribution 4.0 International (CC BY 4.0). It is publicly accessible at:

https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/BMY3UD

B. Dataset Structure

The dataset contains eight files covering both English and French platforms. For this study, two files from the French platform were used:

- `explicit_ratings_fr.tab`: User–item explicit ratings (85,339 rows).
- `items_fr.tab`: Item metadata (1,451 items, 12 columns).

C. Rating Derivation

A key characteristic of this dataset is that ratings are not manually assigned by users. Instead, each rating is derived deterministically from the user's watch percentage: watching $p\%$ of a video yields a rating of $\lfloor p/10 \rfloor$. Watching 100% of a video yields a rating of 10; watching 30% yields a rating of 3. This mechanism produces a highly skewed distribution with a strong peak at rating 10.

TABLE I
RATING FILES USED IN THIS STUDY

File	Rows	Columns
explicit_ratings_fr.tab	85,339	5
items_fr.tab	1,451	12

D. Columns

The `explicit_ratings_fr.tab` file contains: `user_id`, `item_id`, `watch_percentage`, `created_at`, and `rating`. The `items_fr.tab` file contains: `item_id`, `language`, `name`, `nb_views`, `description`, `created_at`, `Difficulty`, `Job`, `Software`, `Theme`, `duration`, and `type`. The `Job`, `Software`, and `Theme` columns were largely empty (stored as empty lists `[]`) in the French dataset, so only `name`, `Difficulty`, and `type` were used for content-based filtering.

III. METHODOLOGY

A. User–Item Matrix Construction

A user–item rating matrix was constructed using pandas `pivot_table` with `user_id` as the row index, `item_id` as the column index, and `rating` as the value. Missing entries were filled with zero. The matrix has dimensions (n_u, n_i) where n_u is the number of unique users and n_i is the number of unique items.

Data sparsity was computed as:

$$\text{sparsity} = 1 - \frac{|\text{ratings}|}{n_u \times n_i} \tag{1}$$

The matrix is highly sparse, as is typical for real-world recommender system datasets. Most users rated only a handful of the 1,451 available items.

B. User-Based Collaborative Filtering: Cosine Similarity

Cosine similarity was computed between all pairs of user rating vectors:

$$\text{sim}_{\text{cos}}(u, v) = \frac{\mathbf{r}_u \cdot \mathbf{r}_v}{\|\mathbf{r}_u\| \|\mathbf{r}_v\|} \tag{2}$$

where $\mathbf{r}_u$ and $\mathbf{r}_v$ are the rating vectors of users u and v respectively, and zeros represent unrated items. The full user similarity matrix was computed in a single call using scikit-learn's `cosine_similarity` function on the user–item matrix.

Recommendations for a target user u were generated by identifying the top-10 most similar users, accumulating a weighted score for each item not yet seen by u:

$$\hat{s}(u, i) = \sum_{v \in \mathcal{N}(u)} \text{sim}(u, v) \cdot r_{vi} \tag{3}$$

where $\mathcal{N}(u)$ is the neighbourhood of similar users and r_{vi} is the rating of user v on item i. Items were ranked by $\hat{s}$ and the top-N were returned.

A fallback mechanism was implemented: if no similar user has rated any unseen item (which occurs when the target user has rated every item in the matrix), the system falls back to globally popular unseen items ranked by total rating count.

C. User-Based Collaborative Filtering: Pearson Similarity

A known limitation of cosine similarity for rating data is its sensitivity to rating scale bias: users who consistently rate everything 9–10 will appear similar to each other purely because of their inflated rating levels, not because of genuine preference alignment. Pearson correlation addresses this by mean-centring each user's ratings before computing similarity.

The Pearson-equivalent similarity was implemented by constructing a mean-centred matrix: for each user u, the mean rating over their rated items is subtracted from each rated entry, while unrated items remain zero:

$$\tilde{r}_{ui} = \begin{cases} r_{ui} - \bar{r}_u & \text{if } r_{ui} > 0 \\ 0 & \text{otherwise} \end{cases} \tag{4}$$

Cosine similarity applied to $\tilde{\mathbf{R}}$ is mathematically equivalent to Pearson correlation. This implementation avoids an expensive pairwise loop over all user pairs and computes the full similarity matrix efficiently via matrix multiplication.

1) Comparison of Cosine and Pearson Similarity: The behavioural difference between the two similarity measures was examined for user 30 (a user who rated exactly 15 items, selected as a representative moderate user). Results:

- **Cosine:** Top-20 similarity scores ranged from 0.30 to 0.42, nearly flat across all 20 neighbours.
- **Pearson:** Top-20 similarity scores ranged from 0.15 to 0.27, with a clear monotonic decrease from the most similar to least similar neighbour.

The flat cosine curve indicates that cosine cannot differentiate between users when the target user gives consistently high ratings, as the vectors are nearly parallel regardless of actual item preference alignment. Pearson's mean-centering removes this bias, producing a more discriminative ranking of neighbours.

D. Content-Based Filtering

Content-based filtering recommends items similar to those a user has previously rated, based on item properties rather than the behaviour of other users. This approach is particularly useful for the cold-start problem where a new user has few interactions.

Item features were combined into a single content string per item:

```
items['content'] = (
  items['Difficulty'].fillna('') + '␣' +
  items['type'].fillna('') + '␣' +
  items['name'].fillna('')
)
```

TF-IDF was applied to the content strings using scikit-learn's `TfidfVectorizer` with English stopwords. An item–item cosine similarity matrix was computed from the resulting TF-IDF vectors. For a query item, the top-N most similar items (excluding the query itself) were returned.

E. Matrix Factorisation via SVD

Matrix factorisation decomposes the user–item rating matrix into latent factor matrices representing user and item embeddings in a shared latent space. This is the approach underlying the Netflix Prize winning solution [3].

1) Mean Centering: Before applying SVD, each user's ratings were mean-centred to remove individual rating scale bias:

$$\tilde{r}_{ui} = r_{ui} - \bar{r}_u \tag{5}$$

where $\bar{r}_u$ is the mean of user u's non-zero ratings. Unrated entries were filled with zero after centering.

2) Truncated SVD: Truncated SVD with $k = 20$ components was applied to the mean-centred training matrix:

$$\tilde{\mathbf{R}} \approx \mathbf{U}_k \mathbf{\Sigma}_k \mathbf{V}_k^T \tag{6}$$

The predicted rating matrix was reconstructed as:

$$\hat{\mathbf{R}} = \mathbf{U}_k \mathbf{\Sigma}_k \mathbf{V}_k^T + \bar{\mathbf{r}} \tag{7}$$

where $\bar{\mathbf{r}}$ is the vector of user mean ratings added back. This gives a full predicted rating for every user–item pair.

3) Train–Test Split: The ratings were split 80%/20% using scikit-learn's `train_test_split` with random state 42. The SVD was fitted exclusively on the training set. Test evaluation was restricted to user–item pairs present in the predicted matrix (i.e., users and items that appeared in the training set).

IV. Results

A. Rating Distribution

The rating distribution is strongly left-skewed, with the majority of ratings concentrated at 10. This occurs because most users who begin a training video tend to watch it to completion, yielding a watch percentage of 100% and thus a rating of 10. Ratings at intermediate values (4–7) are relatively rare.

B. User Activity Distribution

The distribution of ratings per user follows a long-tail pattern. A small number of power users contributed a disproportionately large number of ratings, while most users rated fewer than five items. This sparsity is the primary challenge for collaborative filtering: with only a few ratings per user, finding genuinely similar users is difficult.

C. Collaborative Filtering Results

The cosine-based recommendation function returned no recommendations for the first user in the matrix (user 23) because that user had rated almost every item, leaving no unseen items to recommend from similar users. The fallback to globally popular items was triggered.

The Pearson-based function successfully returned five recommendations for the same user:

- Rédiger un document en dernière minute
- Rejoindre le réseau Yammer
- Qu'est-ce que OneDrive ?
- Introduction à Skype Entreprise
- Créer des groupes de contacts

These are beginner-level Microsoft Office 365 productivity tools, contextually appropriate for a corporate training platform.

D. Content-Based Filtering Results

For item 1 ("Communiquer: qui peut m'aider?"), the content-based system recommended:

- Item 2: Partager : mises à jour régulières, qui peut f... (Difficulty: N/A)
- Item 58: Trouver un modèle de document qui vous convienne (Beginner)
- Item 59: Comparer un document qui a été modifié par que... (Beginner)
- Item 148397: Partager : mises à jour régulières, qui peut f... (Difficulty: N/A)
- Item 390559: Voir qui collabore (Difficulty: N/A)

Two of the five recommended items (58 and 59) are Beginner-level, while the remaining three have no difficulty label in the metadata. The similarity is primarily driven by the item `name` text field, since three of the five results have missing `Difficulty` values. This highlights the limitation of the content model: when `Difficulty` and `type` metadata are absent, TF-IDF relies solely on the item name to compute similarity.

E. SVD Evaluation

The SVD model was evaluated on the filtered test set. Key metrics are summarised in Table II.

1) Interpretation of RMSE: An RMSE of 2.2678 on a 1–10 scale indicates that predicted ratings deviate from actual ratings by approximately 2.3 points on average. For the MARS dataset, this is acceptable given its characteristics: the rating distribution is highly bimodal (ratings cluster at 1 and 10), leaving little signal at intermediate values for the model to learn from. Published benchmarks on comparable sparse e-learning datasets report RMSE values in the range 2.0–3.5 [2].

TABLE II
SVD MODEL EVALUATION RESULTS

Metric	Value
Number of latent components (k)	20
Train/test split ratio	80% / 20%
RMSE (test set)	2.2678
Precision@5	0.0708
Recall@5	0.0179
Rating scale	1 – 10

2) Interpretation of Precision@5 and Recall@5: Precision@5 of 0.0708 means that approximately 1 in 14 recommended items was actually liked (rated $\geq$ 7) by the user. Recall@5 of 0.0179 means that the top-5 recommendations captured 1.79% of all items that user liked in the test set. Both values are low for two structural reasons:

1) **Test set sparsity:** After the 80/20 split, most users have only 1–2 ratings in the test set. Recommending the correct one item from a pool of 1,350 items present in the rating matrix is genuinely difficult.
2) **Training objective mismatch:** SVD was trained to minimise RMSE (a rating prediction objective), whereas Precision@K and Recall@K measure ranking quality. A model optimised for one objective is not necessarily optimal for the other.

These values are within the expected range for sparse implicit/explicit feedback datasets. Published recommender system research on similar corpora reports Precision@5 values of 0.05–0.15 [4].

F. Similarity Score Comparison

TABLE III
COSINE VS. PEARSON SIMILARITY FOR USER 30

Metric	Cosine	Pearson
Top-1 similarity score	0.42	0.27
Rank-20 similarity score	0.30	0.15
Score range (top 20)	0.12	0.12
Shape of decline	Flat	Monotone

Despite having the same absolute range (0.12), the shapes are qualitatively different. The flat cosine curve assigns near-equal weight to all 20 neighbours, diluting the recommendation signal. The monotonically declining Pearson curve concentrates weight on the most genuinely similar users. For a dataset like MARS where many users give predominantly high ratings, Pearson is the more appropriate similarity measure.

V. DISCUSSION

A. Dataset Characteristics and Their Impact

The MARS dataset presents three structural challenges that are uncommon in standard recommender system benchmarks:

1) **Rating skew:** The concentration of ratings at 10 reduces the effective dynamic range available to the model. Mean-centering partially addresses this but cannot recover information that was never in the data.
2) **Extreme sparsity:** Most users rated fewer than 5 of the 1,451 items, leading to a near-empty user–item matrix. Collaborative filtering methods are most effective when users have sufficient rating history.
3) **Domain specificity:** All items are Microsoft Office 365 training videos. This means item content features are highly homogeneous, limiting the discriminative power of content-based filtering beyond difficulty level and video type.

B. Cosine vs. Pearson: Practical Implications

The empirical comparison confirmed the theoretical expectation. In a corporate training context, rating scale bias is particularly pronounced because users may complete videos out of obligation rather than interest, systematically inflating ratings. Pearson's mean-centering effectively removes this bias and makes similarity scores reflect genuine relative preferences rather than absolute rating levels.

C. SVD as Matrix Factorisation

The SVD approach implemented here is a simplified version of the matrix factorisation models that dominated the Netflix Prize competition. The key simplification is the use of scikit-learn's `TruncatedSVD` (which performs a deterministic decomposition) rather than a stochastic gradient descent optimised factorisation model such as FunkSVD or SVD++. The latter incorporate user and item bias terms and are trained to directly minimise the prediction error on observed ratings, typically yielding lower RMSE. The current implementation serves as a conceptual demonstration aligned with the SVD content of the Data Mining curriculum.

D. Content-Based Filtering Limitations

The content-based system relied solely on `Difficulty`, `type`, and `name` because the `Theme`, `Job`, and `Software` metadata columns were empty for most items. This severely limits the feature richness available to the recommender. In a production system, these fields would be populated and would allow much more fine-grained content similarity. The `description` column (subtitle transcripts) was intentionally excluded to keep the content model computationally lightweight.

VI. CONCLUSION

This study implemented and evaluated four recommendation approaches on the MARS French e-learning dataset. The main findings are:

- User-based collaborative filtering with Pearson similarity outperforms cosine similarity on this dataset due to the strong rating scale bias caused by the watch-percentage rating derivation mechanism.
- Content-based filtering produces related item recommendations driven primarily by item name text similarity; the `Difficulty` and `type` metadata fields have limited

impact due to missing values in a significant portion of items.

- Mean-centred SVD with 20 latent components achieved RMSE 2.2678, Precision@5 0.0708, and Recall@5 0.0179 on the test set, consistent with benchmarks for sparse e-learning datasets.
- The low Precision and Recall values are attributable to dataset sparsity and the mismatch between the RMSE training objective and the ranking-based evaluation metrics.

The MARS dataset represents an underexplored domain in recommender systems research. Future work could investigate learning-to-rank objectives (e.g., BPR loss), incorporate the implicit feedback signal (page views from `implicit_ratings_fr.tab`), or experiment with neural collaborative filtering architectures.

Acknowledgment

The authors thank Dr. P.N. Kumar, Faculty, Department of Mathematics, School of Physical Sciences, Amrita Vishwa Vidyapeetham, Coimbatore, for guidance throughout the Data Mining course (Course Code: 24ASD513, Semester 2).

References

[1] M. Farah, A. Hobeika, and J. Issa, "MARS: Mandarine Academy Recommender System Dataset," Harvard Dataverse, 2022. [Online]. Available: https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/BMY3UD

[2] M. Farah, A. Hobeika, and J. Issa, "MARS: A real-world benchmark dataset for recommender systems," *Data in Brief*, Elsevier, 2022.

[3] Y. Koren, R. Bell, and C. Volinsky, "Matrix factorization techniques for recommender systems," *IEEE Computer*, vol. 42, no. 8, pp. 30–37, Aug. 2009.

[4] J. L. Herlocker, J. A. Konstan, L. G. Terveen, and J. T. Riedl, "Evaluating collaborative filtering recommender systems," *ACM Trans. Inf. Syst.*, vol. 22, no. 1, pp. 5–53, Jan. 2004.

[5] P. Resnick, N. Iacovou, M. Suchak, P. Bergstrom, and J. Riedl, "GroupLens: An open architecture for collaborative filtering of netnews," in *Proc. ACM Conf. CSCW*, Chapel Hill, NC, 1994, pp. 175–186.

[6] M. J. Pazzani and D. Billsus, "Content-based recommendation systems," in *The Adaptive Web*, P. Brusilovsky, A. Kobsa, and W. Nejdl, Eds. Berlin: Springer, 2007, pp. 325–341.

[7] G. Salton and C. Buckley, "Term-weighting approaches in automatic text retrieval," *Inf. Process. Manag.*, vol. 24, no. 5, pp. 513–523, 1988.

[8] B. Sarwar, G. Karypis, J. Konstan, and J. Riedl, "Item-based collaborative filtering recommendation algorithms," in *Proc. 10th Int. Conf. World Wide Web (WWW)*, Hong Kong, 2001, pp. 285–295.

[9] J. Han, J. Pei, and M. Kamber, *Data Mining: Concepts and Techniques*, 3rd ed. Elsevier, 2011.

[10] A. Rajaraman and J. D. Ullman, *Mining of Massive Datasets*. Cambridge University Press, 2011.